

Linear Algebra — Exercises

Systems of linear equations

Exercise 1. Find a basis and the dimension of the solution space over the field $\mathbb{R}$ of each of the following systems of linear equations:

$$\begin{cases} x + 2y - 2z + 2s - t = 0 \\ x + 2y - z + 3s - 2t = 0 \\ 2x + 4y - 7z + s + t = 0 \end{cases} \quad \begin{cases} x + 2y - z + 3s - 4t = 0 \\ 2x + 4y - 2z - s + 5t = 0 \\ 2x + 4y - 2z + 4s - 2t = 0 \end{cases}$$

Exercise 2. Discuss the solvability of the following systems of equations over the field $\mathbb{R}$ in terms of the parameters a and b :

$$\begin{cases} ax + by + az + t = 1 \\ ax + y + az + t = 1 \\ x + y + az + at = 1 \end{cases} \quad \begin{cases} x + 2y + 3z + 4t = a \\ 2x + 3y + 4z + 5t = b \\ 3x + 4y + 5z + 6t = a \\ 4x + 5y + 6z + 7t = b \end{cases} \quad \begin{cases} x + ay + az = 1 \\ ax + ay + z = 1 \\ ax + y + az = 1 \\ ax + ay + az = 1 \end{cases}$$

Exercise 3. Find general solutions of the following systems of equations over the field $\mathbb{R}$:

$$\begin{aligned} \text{(a)} \quad & \begin{cases} 2x + 2y + 3z + 4t = 2 \\ 2x + 3y + 4z + 6t = 0 \\ 3x + 4y + 5z + 6t = 1 \\ x + y + z + t = 1 \end{cases} \\ \text{(b)} \quad & \begin{cases} x + y + z - t + u = 1 \\ x - y + z + t + u = 1 \\ -x + y - z + t - u = 0 \end{cases} \\ \text{(c)} \quad & \begin{cases} x + y + z + t + u = 0 \\ x - 2y + 3z - 4t + 5u = 0 \end{cases} \end{aligned}$$

Exercise 4. Find all solutions in $\mathbb{Z}_7$ of the following system of equations:

$$\begin{cases} 3x_1 + 4x_2 + 4x_3 + 6x_4 = 1 \\ 4x_1 + 5x_2 + 4x_3 + x_4 = 0 \\ 5x_1 + 6x_2 + 4x_3 + 3x_4 = 6 \end{cases} \quad \text{over the field } \mathbb{Z}_7.$$